

A County-by-County Analysis of Unemployment, Education, and Rurality in the U.S.

Chelsea Oliveira

Department of Economics at Binghamton University

ECON-383B: Advanced Data Modeling

Dr. Melissa Haller

October 2, 2026

A County-by-County Analysis of Unemployment, Education, and Rurality in the U.S.

Unemployment rates across the U.S. paint a patchwork of economic realities. In some counties, joblessness lingers stubbornly above the national average, while others enjoy persistently lower rates. This contrast is often reduced to a simple rural-urban divide, with policymakers and organizations frequently relying on these assumptions when deciding where to direct resources or plan services; but instead of asking only whether rural counties fare worse, consider: does being classified as rural actually predict a county's unemployment rate once education and income are considered, and does the link between rurality and unemployment shift when these factors are controlled for?

The relationship between geography and labor market outcomes has been documented at both the regional and national level. The Federal Reserve Bank of Richmond has tracked diverging economic trends between rural and urban counties in its regional data reporting, and the Population Reference Bureau has linked low educational attainment to elevated unemployment in the rural Appalachian counties specifically (Federal Reserve Bank of Richmond, 2022; Population Reference Bureau). The U.S. Bureau of Labor Statistics has likewise used county-level data to illustrate the inverse relationship between educational attainment and unemployment nationally (Date, 2026). Using a more recent, broader dataset covering almost all U.S. counties, the subsequent section will test whether rurality alone is a meaningful indicator of unemployment or whether its apparent effect is driven by other, closely related factors. The main hypothesis in this analysis is that county rurality, as measured by the USDA's Rural-Urban Continuum Code, is linked to higher unemployment rates. The null hypothesis is that rurality does not have a relationship with unemployment once other

socioeconomic factors are taken into account. As will be discussed in the results, the answer appears to depend on which variables are included in the model.

Introducing the Data and Methods Utilized

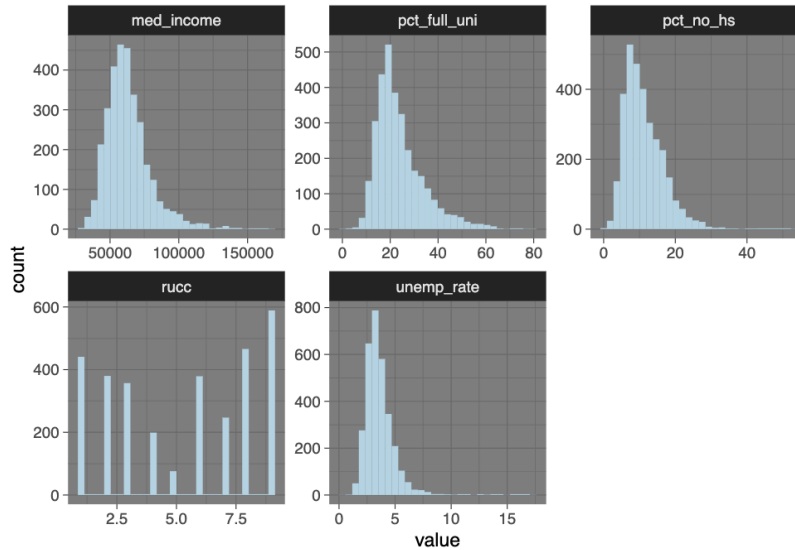
To begin, this analysis uses two county-level datasets published by the Economic Research Service (ERS) of the U.S. Department of Agriculture. The Unemployment and Median Household Income dataset reports annual figures for the civilian labor force, employment, unemployment, and unemployment rate for every U.S. county from 2000–2023, along with the 2022 median household income (Economic Research Service, 2025). The Education dataset reports educational attainment for adults ages 25 and older by county for certain years from 1970 through the 2019–2023 American Community Survey five-year estimates (Economic Research Service, 2025). Both datasets are indexed by Federal Information Processing Standards (FIPS) county codes, which allowed them to be merged into a single analytic file.

Data cleaning was performed in R using the `dplyr`, `readr`, `tidyr`, and `ggplot2` packages from the tidyverse collection (Wickham et. al., 2019). Several steps were necessary before the data were usable for regression. First, both raw files contained aggregate rows for the national total and for each state total, identified by FIPS codes ending in “000”; these were removed to retain only individual counties, since mixing aggregate and county-level observations would have distorted the regression results. Second, several numeric columns, including median household income, were stored as text with comma thousands separators and had to be converted to numeric values before they could be used in calculations. Third, the two cleaned datasets were merged using a left join of the FIPS code, and any county missing a value for one of the key variables was dropped from the final version.

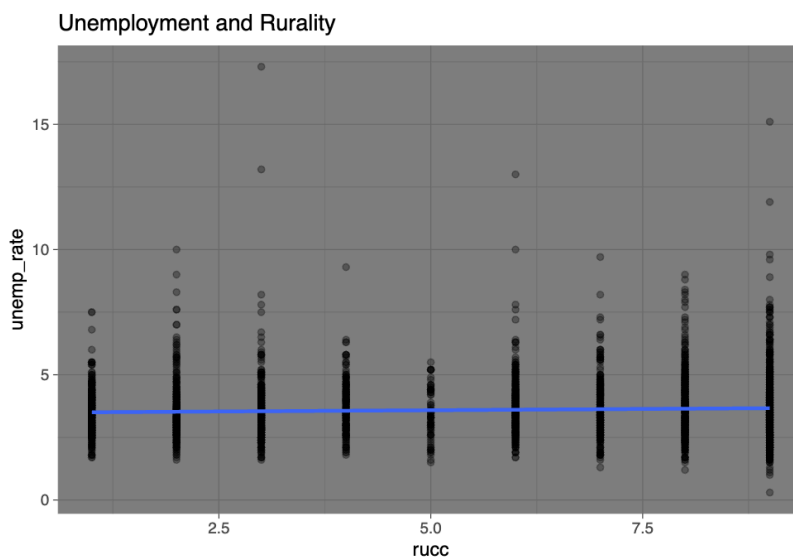
The main variables in this analysis were the 2023 unemployment rate (used as the dependent variable), the 2023 Rural-Urban Continuum Code (a nine-point scale where 1 indicates the most urban counties and 9 the most rural), the percentage of adults aged 25 and older with a bachelor's degree or higher (based on the 2019–2023 estimates), and the 2022 median household income. Three linear regression models were estimated in sequence: the first model regressed unemployment rate on rurality alone, and the second model added educational attainment and median income; a third model using the natural log of the unemployment rate as the dependent variable, estimated after diagnostic testing revealed that the first two models violated key linear regression assumptions.

Descriptive Statistics and Plot Analysis

All four continuous variables in this analysis lean heavily to the right, with most counties bunched at lower values and a handful stretching the distribution far higher. In 2023, the average county unemployment rate was 3.59 percent, but some counties soared as high as 17.3 percent, creating a pronounced right tail. Rurality scored from 1–9 and stood out as the exception; its values were spread more evenly across counties. Meanwhile, the number of adults with a bachelor's degree or higher averaged around 24.0 percent, and median household income hovered around \$63,215. Both displayed the same right-skewed distribution of unemployment.



A scatterplot of unemployment rate versus rurality revealed only a faint upward drift, with data points scattered widely and unevenly across all levels; this visual echoes the weak explanatory power of the first regression model. The pronounced skewness in unemployment rates hints at diagnostic issues ahead, prompting the use of a log transformation later.



Results

Three models were estimated in order. Model 1 included only rurality as a predictor of unemployment rate. Model 2 added educational attainment and median income as additional

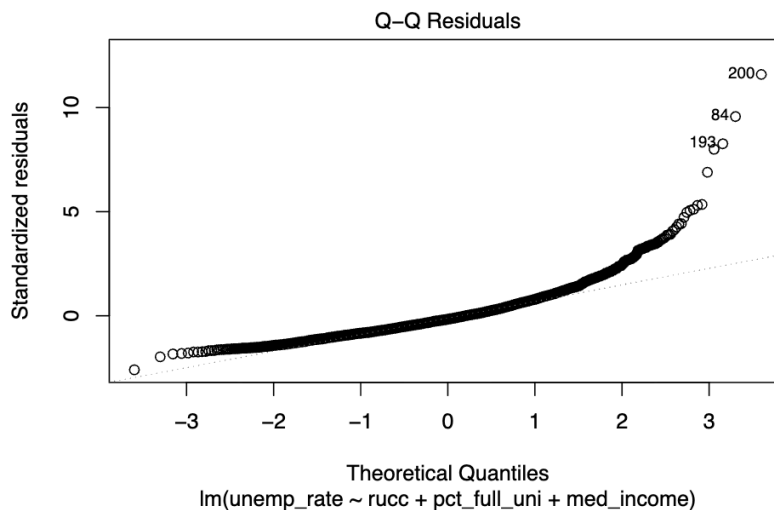
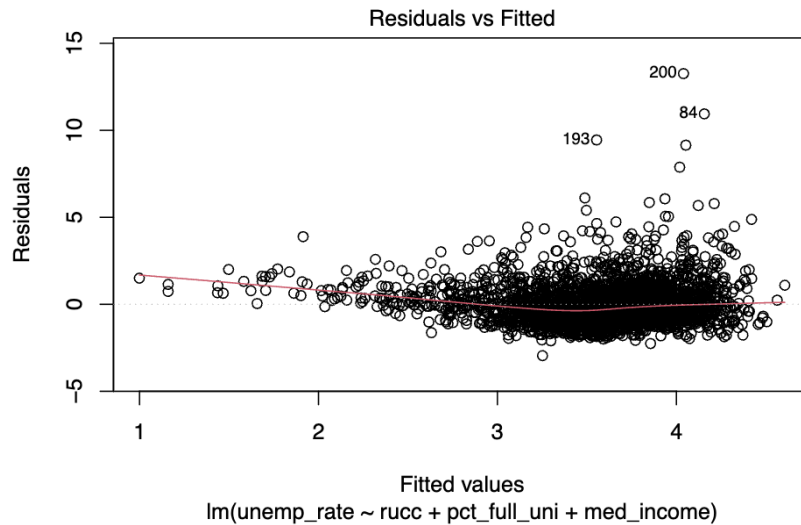
predictors. Model 3 used the same variables as Model 2, but replaced the unemployment rate with its natural logarithm as the dependent variable. This change was made after diagnostic tests suggested that the first two models did not meet the assumptions of homoskedasticity and normally distributed residuals.

Insert model comparisons here.

In Model 1, rurality was statistically significant ($p = 0.0066$) and positive, suggesting that more rural counties have slightly higher unemployment rates; however, the model explained almost none of the variation in unemployment ($R^2 = 0.002$), indicating that rurality alone was a very weak predictor. By contrast, in Model 2, when educational attainment and median household income are added, the rurality coefficient not only flips sign but also becomes strongly significant (coefficient = -0.0604 , $p < 0.001$). This means that after accounting for education and income, more rural counties actually have lower unemployment rates. As expected, both higher education and higher income are linked to less unemployment. Model 2's R^2 climbs to 0.125, a notable jump from Model 1, though much of the variation in county unemployment remains a mystery.

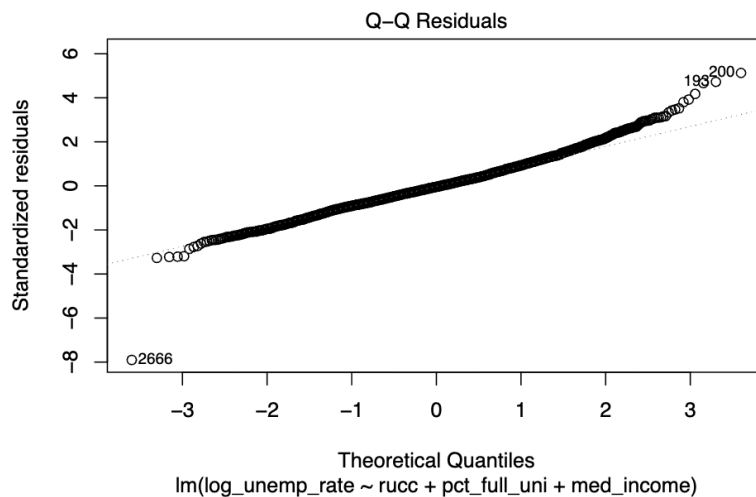
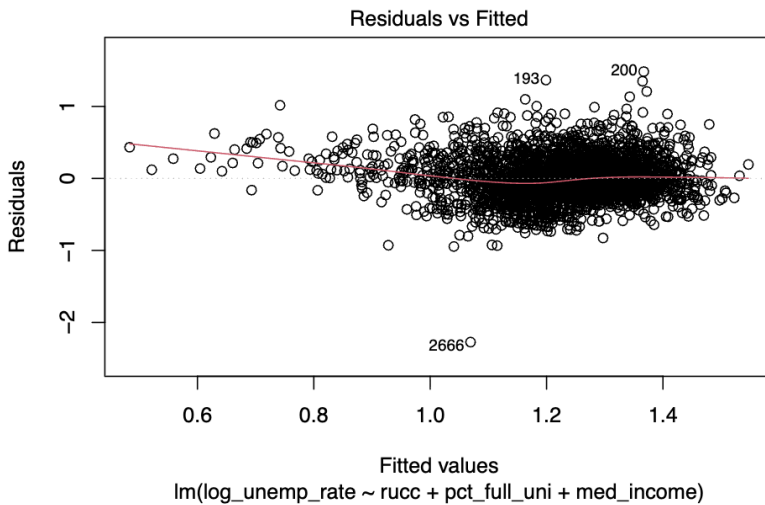
Diagnostic testing of Model 2 revealed problems with the model's underlying assumptions. The residuals-versus-fitted plot shows a funnel shape, with the spread of residuals widening as fitted values increased, indicating heteroskedasticity. The normal Q-Q plot showed residuals departing sharply from the theoretical normal line at the upper tail, indicating non-normally distributed residuals. Both problems are consistent with the right-skewed distribution of unemployment rate observed in the descriptive statistics. A handful of counties, appearing repeatedly across the diagnostic plots as extreme outliers, likely reflect small counties with volatile labor markets rather than a flaw in the model specification itself; none of these

individual counties exceeded the Cook's distance threshold for undue influence on the overall model.



To tackle these issues, the dependent variable was log-transformed, and the model was rerun as Model 3. This adjustment kept all three predictors significant and in the same direction, while boosting the model's fit to an R^2 of 0.155. Residual diagnostics for Model 3 revealed a much tighter, more consistent spread and a Q-Q plot that hugged the normal line more closely through most of the distribution, though a small number of extreme outlier counties remained

even after the transformation. Model 3 is treated as the preferred and final specification for this analysis.



Discussion of Findings and Conclusion

The key takeaway from this analysis is not simply whether rurality relates to unemployment, but that the answer shifts depending on which other factors are considered. Viewed alone, rurality appears to predict higher unemployment. Yet once education and income are factored in, the relationship flips: rural counties actually have somewhat lower

unemployment than urban counties with similar education and income. In this dataset, rural counties tend to have lower educational attainment and income, so these factors may drive the initial relationship seen in Model 1. This finding echoes earlier research on education and unemployment in rural areas like Appalachia and provides a clearer estimate of how much rurality itself matters compared to the socioeconomic factors that often accompany it.

There are several limitations to this analysis. First, the model includes only three predictors, but unemployment is influenced by many other factors that are not included here, such as industry composition, cost of living, local economic shocks, and access to public or private transportation. The relatively low R^2 values in all three models reflect this, since most of the variation in unemployment between counties is likely explained by factors outside the model. Second, this analysis uses data from a single year, so it cannot address trends over time or establish causation, only association. Finally, even after the log transformation, a few counties remained as outliers in all three models, which may indicate that some rural or small-population counties have unemployment patterns that are not well captured by a linear model based on national data.

These results suggest that education and income are more reliable predictors of county-level unemployment than rurality alone. Sweeping claims about rural socioeconomic hardship should be weighed against the socioeconomic realities that often accompany rurality, rather than pinned solely on geography. Additional research could expand on this work by adding industry composition and examining trends over multiple years.

Works Cited

- A grammar of data manipulation*. (2019). Tidyverse.Org. <https://dplyr.tidyverse.org/>
- County-level data sets - education*. (2025). Usda.Gov.
<https://data.ers.usda.gov/reports.aspx?ID=4026>
- County-level data sets - unemployment*. (2025). Usda.Gov.
<https://data.ers.usda.gov/reports.aspx?ID=4038>
- Date. (2026, September). *Education pays, 2025*. Bureau of Labor Statistics.
<https://www.bls.gov/careeroutlook/2026/data-on-display/education-pays.htm>
- Haltom, N. (2022, August 19). *Mapping outcomes across rural and urban communities*. Federal Reserve Bank of Richmond. Regional Matters.
https://www.richmondfed.org/region_communities/regional_data_analysis/regional_matters/2022/rm_08_19_2022_mapping_outcomes
- Multiple (linear) regression in R*. (2024). DataCamp.
<https://www.datacamp.com/doc/r/regression>
- Options* - Yihui Xie | 谢益辉. (n.d.). Yihui.Org. Retrieved <https://yihui.org/knitr/options/>
- Regression Diagnostics in R*. (2025). DataCamp. <https://www.datacamp.com/doc/r/rdiagnostics>
- Scommegna, P. (2012, March 29). *Low education levels and unemployment linked in appalachia - population reference bureau*. Population Reference Bureau.
<https://www.prb.org/news/low-education-levels-and-unemployment-linked-in-appalachia/>
- Wickham, H. (2019). *Create elegant data visualisations using the grammar of graphics*. Tidyverse.Org. <https://ggplot2.tidyverse.org/>